



Course:	Applied Thermodynamics	Course Code:	ME2002
Program:	BS(Electrical Engineering)	Semester:	F 2024
Duration:	60 Minutes	Total Marks:	40
Paper Date:	21-Sep-24	Weight	15%
Section:	ALL	Page(s):	4
Exam:	Mid1 Examination		

Name: _____

Instruction/Notes:

- Use valid assumptions if needed, don't ask any questions.
- Your answers should be correct up to three decimal places with appropriate units.
- Values read from tables should have the same number of significant figures as mentioned in the tables.
- This is an open book, and open notes exam

CLO 1: Analyze a thermodynamics process

Q1 A closed system contains 5 kg of water in a Boiler. Initially, the water has a temperature of 25°C and a specific volume of $1.25616 \text{ m}^3/\text{kg}$. The system operates in two stages: [20]

- Stage 1: The system is heated until the pressure inside the boiler reaches 200 kPa keeping volume constant.
- Stage 2: The system is then cooled from a temperature of 150°C , allowing the pressure and temperature to change, but keeping volume constant.

a) Describe the phase and conditions of the water at $v = 1.25616 \text{ m}^3/\text{kg}$ and $T = 25^{\circ}\text{C}$?

b) When the pressure inside the boiler reaches 200 kPa, what will be the state of the water? Determine the corresponding temperature.

c) After the system cools from 150°C , what will be the total volume V of the water in the system? Describe the state of the water at this point.

d) If the water is a saturated mixture of vapor and liquid, what is its quality at $T = 100^{\circ}\text{C}$? Also, illustrate this state and quality on a $T-v$ diagram.

CLO 1: Analyze a thermodynamics process

Q2 Urea plants use ammonia as a working substance. In such a set up Ammonia is at 10°C with mass of 300 grams in a piston cylinder with an initial volume of 0.03 m^3 . The piston has a design such that a pressure of 900 kPa will float it. Now the ammonia is slowly heated to 50°C . Identify the states involved, find the final pressure and volume and draw this entire process on a Pv diagram. [20]

0.3 kg

**Applied Thermodynamics
(ME2002)**

Date: November 2, 2024

Course Instructor(s)

Dr Huzaifa Rauf

Mr. Muhammad Abdul Majid

Sessional-2 Exam

Total Time: 1 Hours

Total Marks: 40

Total Questions: 02

Semester: F-2024

Campus: Lahore

Deptt: Electrical
Engineering

Student Name

Roll No

Section

Student

Vetted by

Vetter Signature

Instruction/Notes:

- Submit the question paper along with your answer sheet.
- Use valid assumptions if needed, don't ask any questions.
- Your answers should be correct up to three decimal places with appropriate units.
- Values read from tables should have the same number of significant figures as mentioned in the tables.
- You are allowed to bring two A4 sized cheat sheets, as well as thermodynamic tables

CLO #2: Analyze energy of a system using energy equation

Q1:

[marks 20] / 20

A steam turbine receives steam at its inlet at 3 Kg/s, 1000 Kpa, 400°C, at a height of 6m, and 40 m/s. The heat loss is 10KJ/Kg. Exit is at 100°C, at a height of 1m and 5 m/s as saturated vapor. Find out power produced and specific work done.

Now if the condenser at the output of the turbine improves its performance such that the exit state becomes 50°C, and 5 m/s as saturated vapor, keeping everything else constant. How much additional power is produced in the second case?

CLO #3: Analyze modes of heat transfer

Q2:

[marks =20]

20 / 11

An electric water-heater has the shape of a perfect cylinder. The heating rods are present inside the heater. All the surface area of heater is covered up with insulation boards. The diameter of the water heater (excluding the insulation) is 0.2m and its length is 1.5m. The inner insulation surface is at 80°C and the outside surface is at 25°C and the board material has a conductivity of 0.09 W/m K. Find out thickness of insulation board required to limit the heat transfer loss to 200 W?

Assuming that the outside air temperature is 9°C find out heat lost through radiation. Also find out two different values of co efficient of convection ignoring heat radiation and not ignoring heat radiation.

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